NOTEBOOK 02C

Python for Actuaries Part 2

Agenda

In this notebook, we will cover:

- File operations
  - Opening files
  - Reading files
  - Writing files

File Operations in Python

In Python, we can easily work with files, whether for reading data or writing information. Working with files involves three steps:

1. **Opening** the file: We can open a file using the `open()` function to read or write.
2. **Reading or Writing**: Once the file is opened, we can perform the desired operations.
3. **Closing** the file: After the operations are completed, the file is closed to free up resources.

Basic File Operations:

- **Reading a file**: We can read the contents of a file using `read()` or `readline()`.
- **Writing to a file**: We can write text to a file using `write()`.
- **Handling file paths**: Absolute and relative paths can be used to save or open files in different locations.

Code Examples:

1. Reading a File

In this example, the contents of a text file are read in their entirety.

```
# Opening a file in read mode ('r')
with open('example.txt', 'r') as file:
    content = file.read()  # Read the entire content
    print(content)
```

2. Writing to a File

To write text to a file, we open the file in write mode 'w'.

```
# Opening a file in write mode ('w')
pass
```

3. Reading a File Line by Line

Sometimes it is more efficient to read the file line by line.

```
# opening a file in read mode ('r')
pass
```

4. Handling File Paths

We can also write and read files in locations other than the current folder:

```
# writing to a file in folder
pass
```

```
# reading from a file in folder
pass
```